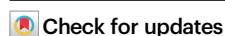


Decoding the substrate specificity landscape of a promiscuous enzyme through multi-substrate mutational scanning

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Rosario Vanella ^{1,2}✉, Sean Boulton^{1,2,3}, Christoph Küng ^{1,2} & Michael A. Nash ^{1,2,4,5}✉

Substrate specificity is a defining feature of enzyme function, but its molecular underpinnings remain difficult to decode and engineer. Here, we leverage enzyme proximity sequencing (EP-Seq) to systematically map how single-point and combinatorial mutations reshape the substrate preferences of D-amino acid oxidase (DAOx) from *Rhodotorula gracilis*, a model promiscuous enzyme. We generate ~40,000 sequence–phenotype pairs, enabling us to profile the activities of ~6,500 unique DAOx variants against five D-amino acid substrates with distinct physicochemical properties. Our analysis reveals that substrate-specific mutations are distributed throughout the enzyme structure. Mutations near the active site drive strong specificity shifts but also incur catalytic penalties, while distal mutations subtly rewire intramolecular contacts in order to modulate specificity with minimal loss of activity. We identify and validate positional hotspots that act allosterically to influence specificity, and characterize key variants that acquire exclusive substrate specificity or exhibit up to 230-fold changes in substrate preference. Combining mutations with complementary effects further sharpens substrate discrimination, enabling rational design of highly selective biocatalysts. This work establishes a powerful framework for decoding enzyme specificity and provides foundational datasets to advance AI-guided enzyme engineering.

Tailoring substrate specificity in enzymes is crucial for biocatalyst engineering, therapeutic enzyme design, and synthetic biology. Despite major progress in structural and computational enzymology, our understanding of how enzymes discriminate between structurally similar substrates remains incomplete, particularly in enzymes that exhibit broad promiscuity. Substrate specificity was for many years considered a static property¹, but is now appreciated as an emergent feature shaped by an interplay of active-site geometry, conformational dynamics, and long-range intramolecular interaction networks, all under complex evolutionary constraints^{2–4}.

Substrate specificity manifests as a relative property, enabling enzymes to preferentially convert their primary substrate relative to

structurally similar alternatives, while maintaining sufficient reaction flux in the crowded cellular environment^{3,5,6}. However, the molecular principles governing substrate discrimination and its co-evolution with other enzyme properties (e.g., overall catalytic activity and folding stability) remain poorly understood. This knowledge gap limits our ability to predict how mutations will affect enzyme performance and impedes the rational engineering of enzymes for diverse applications.

Deep mutational scanning (DMS) has revolutionized the study of protein function by enabling high-throughput mapping of sequence–fitness relationships^{7–11}. If successfully applied to the study of enzyme specificity, DMS could have significant potential to illuminate how sequence encodes discrimination between structurally similar

¹Department of Chemistry, University of Basel, Basel, Switzerland. ²Department of Biosystems Science and Engineering, ETH Zurich, Basel, Switzerland.

³Department of Health Sciences and Technology, ETH Zürich, Zürich, Switzerland. ⁴National Center for Competence in Research (NCCR), Molecular Systems Engineering, Basel, Switzerland. ⁵Swiss Nanoscience Institute, Basel, Switzerland. ✉e-mail: rosario.vanella@unibas.ch; michael.nash@unibas.ch

substrates. However, applying DMS to enzymes presents challenges. DMS workflows applied to enzymes must directly link catalytic activity to a measurable cellular phenotype. This requirement has largely limited prior work to enzymes whose activity can be directly linked with cell growth or survival^{12–14}. Prior DMS work on substrate specificity has therefore focused on antibiotic-resistant β -lactamase^{15–17} and on an amide hydrolase¹⁸ whose function generates a nitrogen source required for cell proliferation.

These studies demonstrate that each substrate shapes the enzyme's fitness landscape in a distinct way, and offer preliminary insights into molecular determinants of specificity. However, this line of research has been limited by reliance on intracellular reactions and growth-based selections, where reaction conditions may be poorly controlled, substrate concentrations cannot be precisely set, and the observed phenotypic fitness may conflate the influence of mutations on catalytic activity with their impact on enzyme abundance (i.e., expression level) or folding stability.

To overcome these challenges, we recently developed enzyme proximity sequencing (EP-Seq), a DMS workflow that leverages a high-throughput single-cell enzyme activity assay based on radical-mediated fluorescent proximity labeling^{19–24}. EP-Seq enables quantification of enzyme activity for cells displaying an enzyme variant on their surface, where catalytic turnover can be linked, directly or indirectly via a cascade reaction, to the production of hydrogen peroxide (H_2O_2). We previously validated EP-Seq by mapping the mutational landscape of an oxidoreductase¹⁹, showing that the method can disentangle the effects of mutations on expression level, folding stability, and catalytic activity in a massively parallel and quantitative manner.

In the present work, we extend the use of EP-Seq to examine how mutations affect substrate selectivity in a D-amino acid oxidase from *Rhodotorula gracilis*, a naturally promiscuous enzyme with diverse industrial and biomedical applications^{25–28}. By displaying a comprehensive library of single-substitution enzyme variants on the surface of yeast cells, we measured the activity of thousands of variants against five substrates with distinct biophysical properties. This setup allowed precise control over reaction conditions and facilitated direct comparison of fitness scores across datasets.

By analyzing the fitness landscape of DAOx on five substrates, we identified hundreds of single amino acid substitutions that differentially modulate enzyme performance, revealing how mutations that control substrate preference are distributed across the entire protein structure. Mutations near the active site produced large specificity shifts but often incurred catalytic penalties, while distal substitutions more subtly fine-tuned substrate preferences with minimal loss of activity. This highlights distinct mechanisms for achieving specificity depending on the extent of functional cost that can be tolerated during evolution. We further developed a substrate specificity score and validated this statistical metric using 140 single-clone measurements of enzymatic discrimination factors. This high-throughput DMS-based phenotypic metric enables rapid and quantitative assessment of enzyme selectivity between substrate pairs. Finally, we showed that combining individually beneficial mutations can synergistically sharpen substrate specificity, offering a scalable route to engineering highly selective biocatalysts. Together, these findings establish EP-Seq as a robust platform for decoding and engineering enzyme specificity at scale.

Results

EP-Seq workflow applied to study substrate-specific effects of mutations

To systematically map how mutations affect substrate specificity in *Rhodotorula gracilis* D-amino acid oxidase (DAOx) we applied EP-Seq¹⁹, a high-throughput platform that links enzyme activity to a fluorescence readout at the single-cell level. In this workflow, a library of enzyme variants fused to the Aga2 anchor protein is displayed on the surface of *Saccharomyces cerevisiae*. Enzymatic turnover is coupled to

production of H_2O_2 , which in the presence of horseradish peroxidase (HRP) initiates a tyramide-based radical labeling reaction that links fluorophores to the cell surface. The resulting cellular fluorescence intensity correlates with the catalytic turnover of the variant borne by that particular cell. Following labeling, cells are sorted by fluorescence-activated cell sorting (FACS) into bins, and next-generation sequencing (NGS) analysis of the DNA barcodes reveals the identities of the enzyme variants in each bin. Fitness scores for each variant are then computed by comparing their bin distributions to that of the wild-type enzyme, allowing quantitative measurements of activity. In a parallel experiment, the expression levels of all variants in the library are monitored by fluorescently labeling a terminal epitope tag (e.g., 6 $\times$ His tag) on the displayed enzyme construct, and again quantifying the expression fitness by FACS and NGS. By decoupling activity and expression measurements, EP-Seq enables the construction of orthogonal fitness landscapes for catalytic activity and display levels.

We applied EP-Seq to evaluate the effects of single-point mutations on the catalytic activity of DAOx across five distinct D-amino acid substrates (Fig. 1A). The substrate panel included three hydrophobic amino acids (D-alanine (D-Ala), D-methionine (D-Met), and D-phenylalanine (D-Phe)) with progressively larger side chains (methyl, S-methyl thioether, and phenyl, respectively). Additionally, we included two hydrophilic substrates (D-asparagine (D-Asn), and D-glutamine (D-Gln)) bearing polar amide groups on two- and three-carbon side chains, respectively.

We first cloned the wild-type DAOx (WT-DAOx) gene in-frame with Aga2, and confirmed that the surface-displayed fusion protein on *S. cerevisiae* strain EBY100 retained activity and substrate preferences comparable to the soluble enzyme. We found that the hydrophobic substrates were processed with higher affinity than the polar ones (Supplementary Note 1, Supplementary Fig. 1, Supplementary Table 1). For the EP-Seq assays, each substrate was assayed at its experimentally determined K_M concentration. Time-course optimization of the tyramide reaction showed that a 60-min reaction time yielded optimal labeling for the hydrophobic substrates, while 120 min were required for the less reactive polar substrates. These conditions ensured comparable signal intensities and dynamic ranges across all five substrates (Supplementary Note 2, Supplementary Fig. 2).

We next transformed EBY100 yeast cells with a site-saturation mutagenesis library of DAOx variants, covering 94.4% of all possible single amino acid substitutions (6530 of 6916 possible missense variants; Fig. 1B). After the tyramide labeling assay, cells were sorted by fluorescence intensity. A low fluorescence gate captured cells either not displaying the enzyme or bearing inactive variants, while the remaining cells were equally divided into three bins of increasing signal (Fig. 1C, Supplementary Fig. 3). Barcode sequencing of each bin was then used to reveal the identity and frequency of the variants, from which we computed an activity fitness score for each variant relative to WT (Fig. 1D, E, Supplementary Table 2, Eqs. (1–3)).

Validation of the EP-Seq workflow and analysis of fitness scores

To assess the reproducibility of the workflow, we calculated Pearson correlation coefficients (r values) between fitness scores obtained from two biological replicates. This analysis included DNA single-mutant variants represented by at least 10 cells per replicate at the screening stage ($n = 6573$). Across all five substrates, Pearson's r values ranged from 0.94 to 0.97, indicating high reproducibility of the assay (Fig. 2A).

Within each screen, we computed a consensus score for each DAOx variant (Methods, Eq. (4)), and examined the distributions of scores for missense, nonsense and synonymous mutations (Fig. 2B, C). Missense mutations were generally associated with reduced catalytic activity (average score = -0.31 ± 0.05 ; $n = 5821$ variants in all five datasets) (Fig. 2B, C, Supplementary Table 3). Nonsense mutations had strongly negative effect on activity (average score -0.58 ± 0.05 ; $n = 316$

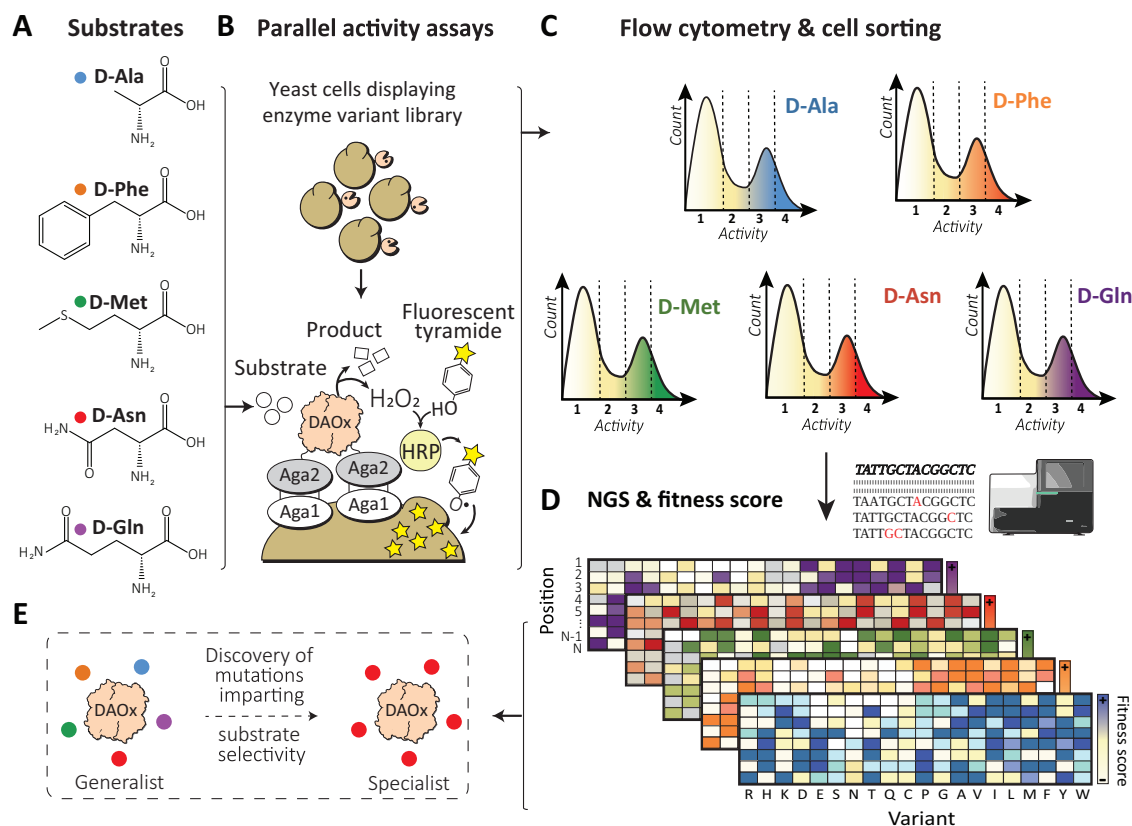


Fig. 1 | EP-Seq workflow for mapping substrate specificity in the promiscuous enzyme D-amino acid oxidase (DAOx). **A** Five D-amino acid substrates with distinct physicochemical properties were selected to profile the activity of a comprehensive DAOx variant library. **B** Yeast cells display DAOx variants on their surface. Enzyme activity is coupled to H_2O_2 production, which initiates a tyramide-based labeling reaction catalyzed by horseradish peroxidase (HRP). This covalently attaches fluorescent radicals onto cells displaying active variants, linking catalytic activity to fluorescence intensity. **C** Each substrate assay is analyzed by flow

cytometry and fluorescence-activated cell sorting (FACS), partitioning cells into four activity-based bins. **D** Barcode sequencing of the sorted populations reveals the identity and abundance of variants in each bin, allowing computation of a fitness score that quantifies activity on each substrate relative to WT. **E** Comparative analysis across substrates identifies specificity-determining mutations, reveals structural hotspots for engineering selective variants, and provides insights into the biophysical trade-offs between activity, specificity and stability.

in all five datasets), indicating that loss of the C-terminal region beginning anywhere from position 1 to position 354 had a deleterious effect on the catalytic activity of the enzyme, regardless of the substrate used (Fig. 2C, Supplementary Table 3). In contrast, variants carrying premature stop codons beyond position 356, which resulted in loss of only the last -10 amino acids, exhibited fitness scores comparable to WT (average score -0.01 ± 0.03 ; $n = 9$ variants in all five datasets) (Fig. 2C), suggesting that this C-terminal region is largely unnecessary for catalytic function, regardless of the substrate. Synonymous mutations did not substantially alter fitness, with scores similar to WT (avg score $= 0.00 \pm 0.0035$; $n = 281$ in all five datasets) (Fig. 2B, C, Supplementary Table 3).

To validate the fitness scores obtained from EP-Seq, we used an Amplex Red assay to measure the catalytic activity of 18 yeast-displayed single-mutant DAOx variants on each of the five substrates. Initial reaction rates were normalized to that of the WT enzyme assayed in parallel and compared to the corresponding DMS-derived fitness scores. The two measurements were highly consistent, with Pearson's r values ranging from 0.93 to 0.98 across all tested conditions (Fig. 2D).

Comparative analysis of EP-Seq scores across five DAOx substrates

We then combined the five high-quality EP-Seq datasets. The resulting data comprised 6418 unique variants, including 5821 missense, 316

nonsense, and 281 synonymous mutations. To account for differences in enzyme display (i.e., expression) levels, activity scores were normalized to expression scores obtained from our previously published dataset⁹. We further only considered variants with expression levels higher than the mean expression of nonsense variants plus one standard deviation. Synonymous variants were used to define a reference range for WT-like behavior (average score across all datasets: 1.00 ± 0.037). Nonsense and synonymous mutations were subsequently excluded, leaving 5800 missense variants (83.9% of the total) shared across all five substrate screens that were used for comparative downstream analysis.

To qualitatively compare the effects of mutations, we visualized the fitness scores using heatmaps, which revealed broadly similar activity landscapes across all five screens (Supplementary Fig. 4). Quantitative analysis further confirmed strong concordance in these datasets, with Pearson's r values ranging from 0.93 to 0.96 in pairwise comparisons between different substrates (Supplementary Fig. 5). The weakest correlation ($r = 0.93$) was observed between D-Ala and D-Gln (i.e., the smallest hydrophobic and the largest polar substrates, respectively). In contrast, the strongest correlations ($r = 0.96$) were found between D-Met and each of D-Phe, D-Ala, D-Asn, highlighting the intermediate size and chemical features of D-Met in comparison to the other substrates tested.

We attribute the high level of agreement of the five datasets to the inherently promiscuous and generalist nature of *R. gracilis* DAOx, a

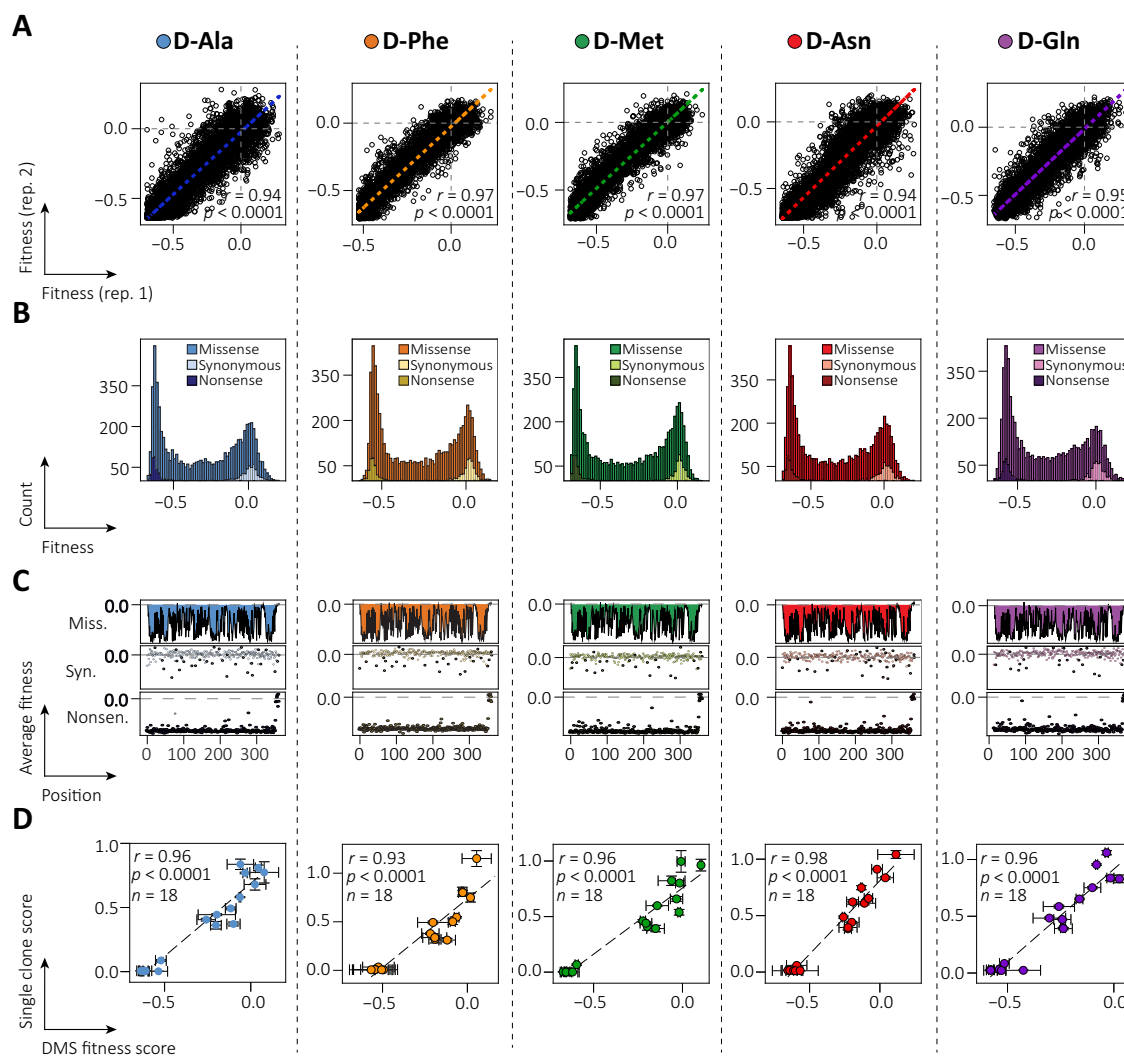


Fig. 2 | Validation of EP-Seq fitness scores quantifying DAOx activity across five substrates. A Scatter plots and linear regression of activity fitness scores calculated from two biological replicates for each of the five substrate screenings ($n = 6573$). **B** Distribution of activity fitness scores for missense, synonymous, and nonsense mutations across all variants and substrates. **C** Average fitness effect per sequence position for missense (top), synonymous (middle) and nonsense

(bottom) mutations. **D** Correlation between DMS-derived fitness scores and catalytic activity measured independently in monogenic yeast culture for 18 single DAOx variants. Dashed line: linear regression; r : Pearson correlation coefficients, n : number of datapoints, p : p -values two-tailed. Source data are provided as a Source Data file.

trait that is also observed in more recently diverged eukaryotic homologs such as human DAOx. Additionally, our use of standardized assay conditions (i.e., K_M -matched substrate concentrations and optimized timepoints) facilitated a consistent dynamic range across all substrate screens, despite significant substrate-dependent differences in catalytic efficiency. This minimized potential detection biases and enhanced scoring accuracy, particularly for low-activity variants where substrate-specific effects are most likely to occur due to functional trade-offs in enzyme evolution^{3,15,19}.

Identification of specificity-determining positions in DAOx

To identify mutations that alter substrate preference, we computed z-scores for each missense variant across all pairwise combinations of the five substrate datasets. These z-scores quantify how much a variant's fitness difference between two substrates deviates from the average difference observed across all mutations between the same two substrates. A positive z-score indicates enhanced activity on one substrate relative to its paired counterpart, while a negative z-score indicates the opposite (Supplementary Fig. 6). We refer to this metric as the substrate specificity score (Sp-score), which captures both the

magnitude and direction of a mutation's impact on substrate specificity (Methods, Eq. (5)).

To define a significance threshold, we used synonymous mutations as an internal control population to estimate background noise. Mutations with $|\text{Sp-score}| \geq 3$, corresponding to three standard deviations from the mean, were classified as substrate-specific mutations. This threshold excluded >99% of synonymous variants and retained 601 missense mutations (332 unique variants) that were then classified as substrate-specific (Fig. 3A, B, Supplementary Fig. 6). These mutations were broadly distributed throughout the DAOx sequence, affecting 166 of 364 positions (~45%) (Fig. 3A).

When mapped onto the 3D structure, substrate-specific mutations appeared throughout the enzyme, with an average distance of 18–24 Å from the catalytic center depending on the substrate. However, mutations ≤ 8 Å from the active site tended to show the strongest effects (i.e., highest $|\text{Sp-score}|$), highlighting the important contribution of active-site and first shell residues to specificity (Fig. 3C).

While most specificity-determining positions were unique to one or two substrates, a small set showed broader effects. We

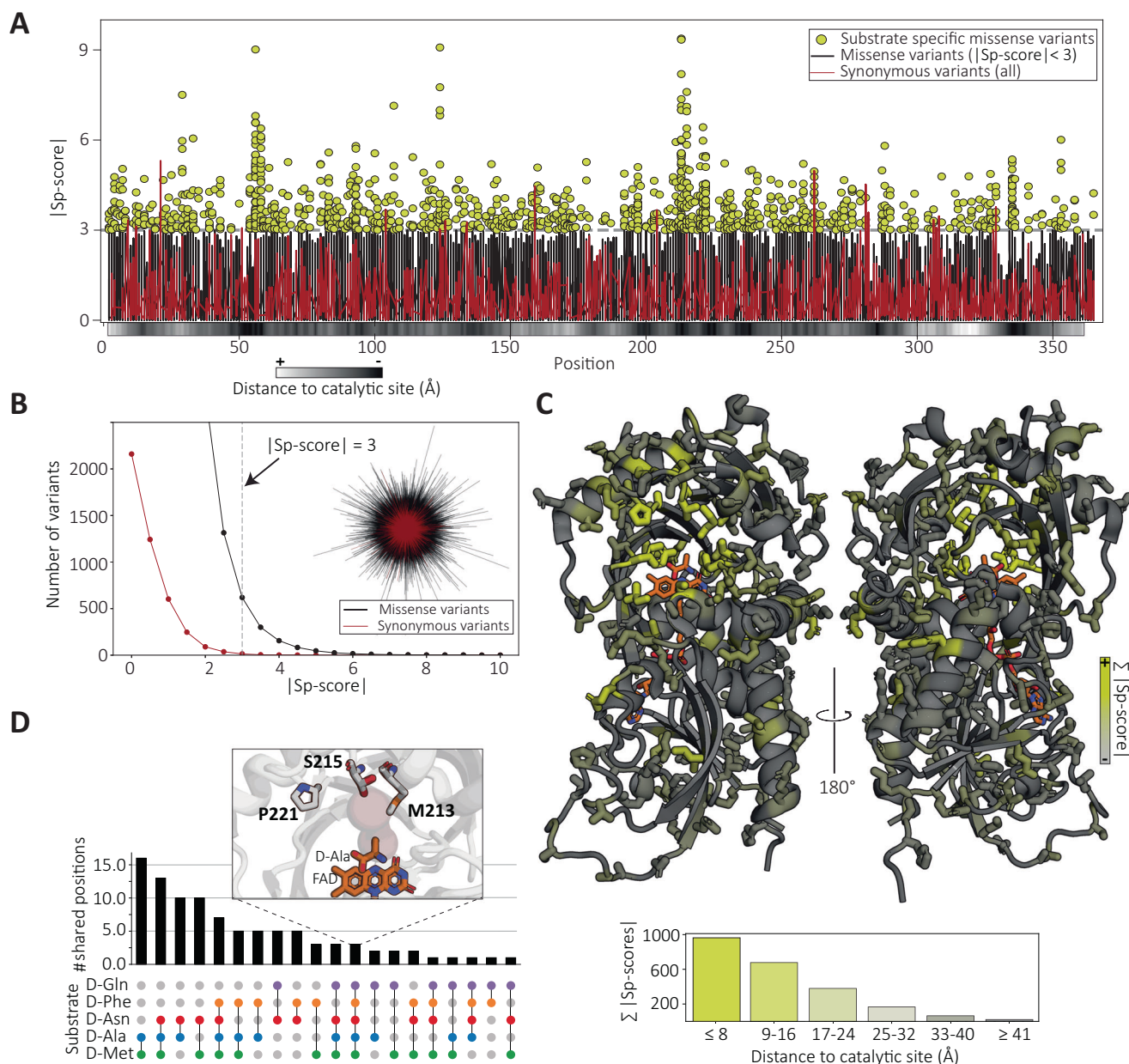


Fig. 3 | Identification and structural mapping of DAOx mutations conferring substrate specificity. **A** Scatter plot of absolute specificity scores across DAOx positions for all missense and synonymous variants and for each of the 10 substrate pairs. Missense variants with $|Sp\text{-score}| \geq 3$ are shown as yellow dots. The x-axis color gradient indicates the distance of each position to the catalytic site. **B** Relationship between the number of variants and their corresponding absolute z-scores. A threshold of $|Sp\text{-score}| \geq 3$ was selected to include $<1\%$ of synonymous mutations. Inset: starburst plot comparing z-score distributions of missense and synonymous variants, illustrating the clustering of synonymous mutations near zero Sp-score. Sp-score increases radially outward from the center. **C** (Top) DAOx crystal structure (PDB ID: 1COP) with substrate-specific positions highlighted as sticks. Color

intensity (gray to yellow) represents the cumulative absolute Sp-score per position, considering only Sp-scores >3 . This reflects the contribution of each site to the global substrate-specificity landscape. The FAD cofactor and D-Ala substrate found in the catalytic site of the enzyme are shown as orange sticks. Bottom: bar plot showing the relationship between the distance of substrate-specific mutations from the catalytic site and their cumulative absolute specificity scores. **D** UpSet plot showing unique and shared DAOx positions with substrate-specific effects across the five tested substrates. Inset: zoom-in on positions 213, 215, and 221, which alter substrate specificity across all five substrates depending on the substituting residue (see Supplementary Fig. 7). Source data are provided as a Source Data file.

found three positions (residues 213, 215, and 221) that exhibited differential substrate-specific effects across all five tested substrates, depending on the identity of the introduced amino acid (Fig. 3D, Supplementary Fig. 7). The greatest overlap in substrate-specific positions was observed between D-Ala and D-Met ($n = 15$), suggesting a convergent mutational pattern shaped by their similar physicochemical profiles. In contrast, D-Phe and D-Gln shared only one specificity-associated site despite being similarly

bulky. This limited overlap highlights the influence of chemical composition over steric similarity in shaping the specificity landscape (Fig. 3D).

Validation and global analysis of substrate specificity scores

To validate the EP-Seq-derived substrate specificity scores (Sp-scores), we performed kinetic measurements on 17 single-mutant DAOx variants exhibiting high Sp-scores towards at least one of the five

substrates. Each variant was displayed on yeast, and Michaelis–Menten kinetics were measured on all five substrates. For each variant and substrate pair, we then calculated discrimination factors (D-factors). The D-factor, defined as $(k_{\text{cat}}/K_{\text{M}})_{\text{sub1}} / (k_{\text{cat}}/K_{\text{M}})_{\text{sub2}}$ (Methods, Eq. (8)) provides a standardized metric of specificity, representing catalytic efficiency on a cognate substrate divided by that on a non-cognate substrate^{3,6}. We then compared Sp-scores with D-factors across 140 variant-substrate combinations. The two measures were strongly correlated (Pearson's $r = 0.76$, $p < 0.0001$, $n = 140$, Fig. 4A), confirming that the EP-Seq-derived Sp-scores accurately reflect substrate-selective catalytic behavior.

We next analyzed the distribution of substrate-specific mutations across all five assays. Among 445 unique single-mutant variants with $|\text{Sp-score}| \geq 3$ towards at least one of the tested substrates, the majority favored D-Met ($n = 121$), D-Asn ($n = 120$), or D-Ala ($n = 110$) (Fig. 4B, pie chart). The ten most selective variants for the smallest two substrates D-Ala or D-Asn (irrespective of the non-cognate substrate) exhibited significantly higher average Sp-scores (D-Asn: 7.91 ± 1.10 , $n = 10$; D-Ala: 6.28 ± 0.54 , $n = 10$) as compared to the top ten most specific variants for the three larger substrates, including hydrophobic D-Phe (4.87 ± 0.53 , $n = 10$) and D-Met (5.59 ± 1.00 , $n = 10$), or the polar D-Gln (5.45 ± 0.88 , $n = 10$) (Fig. 4C). These trends suggest that the strongest substrate-selective effects arise when mutations act to exclude sterically larger non-cognate substrates.

To quantify patterns of substrate discrimination, we next calculated the mean Sp-score values for each substrate pair by multiplying the number of selective variants ($|\text{Sp-score}| \geq 3$) for a given substrate pair by their average Sp-score, and normalizing by the total number of variants selective for the same primary substrate. This produced a single value reflecting both the frequency and intensity of discrimination across all substrate pairs (Fig. 4B, bars). Discrimination was most pronounced when substrate pairs differed substantially in terms of size or polarity. For example, D-Ala- and D-Asn-specific variants most effectively excluded the bulkier substrates D-Phe and D-Gln. Conversely, D-Phe-specific variants discriminated strongly against D-Gln but less so against the smaller substrates D-Ala, D-Asn, and D-Met. A similar pattern held for D-Gln-specific variants, which discriminated against D-Phe but not against smaller substrates. These trends suggest that accommodating the larger substrates requires an open active site that also admits smaller substrates, limiting selectivity based on steric exclusion. The strong mutual discrimination between D-Phe and D-Gln likely reflects changes in electrostatic interactions. Finally, D-Met-specific variants exhibit moderate and balanced selectivity across all four other substrates, consistent with its intermediate size and chemical features (Fig. 4B, bars).

A detailed analysis of the amino acid substitutions in the most selective variants revealed distinct trends, especially for D-Ala and D-Asn. Among the top 10 variants selective for D-Ala, 7 introduced residues bulkier than the WT residue, typically tyrosine (Y) or tryptophan (W) at positions near the catalytic site (average distance = $4.62 \text{ \AA} \pm 2.54$; Supplementary Table 4). This suggests that steric exclusion of larger non-cognate substrates is a dominant mechanism for enhancing selectivity on D-Ala. In contrast, only 2 of the top 10 D-Asn-selective variants showed increased bulkiness while 8 introduced charged or polar side chains (Supplementary Table 4), suggesting that substrate exclusion alone was insufficient to generate selectivity on D-Asn, which the wild-type enzyme poorly accommodates ($K_{\text{M}} > 10 \text{ mM}$).

Analysis of the 10 most selective variants on D-Phe, D-Met, and D-Gln, which exhibited lower average Sp-scores than the 10 most selective variants on D-Ala or D-Asn, did not reveal any consistent substitution trends (Supplementary Table 4), suggesting that specificity for these substrates arises through more complex or context-dependent mechanisms.

Together, these results indicate that steric clashes and electrostatic interactions play dominant roles in driving substrate specificity.

This observation supports the broader idea that positive selection (i.e., enhancing activity on a cognate substrate) is typically a slow and incremental evolutionary process. By contrast, specificity can emerge more rapidly through negative selection, where mutations exclude non-cognate substrates via structural or electrostatic incompatibility^{3,4,29,30}. This strategy also applies in many laboratory evolution campaigns, where undesired activity is penalized while variants enhancing specificity are often reached by excluding secondary substrates from the catalytic site^{31–33}.

Evolutionary interplay between substrate specificity, catalytic activity and structural stability

Engineering enzymes for improved activity often compromises structural stability, a well-established trade-off reflecting the delicate balance between function and fold^{19,34–37}. To investigate how this balance shifts during evolution of substrate specificity, we analyzed substrate-specific DAOx variants by combining the catalytic activity data with yeast display expression scores, known to be a valid proxy for folding and stability^{19,38–40}.

Among the 445 specificity-conferring single mutants, 24.9% ($n = 111$) retained catalytic activity toward the primary substrate comparable to WT. This suggests that specificity can emerge through functionally neutral drift. These mutations were mildly destabilizing (average expression score: 0.95 ± 0.08 ; WT score: 1) and generally located away from the active site (average distance: $13.60 \pm 10.29 \text{ \AA}$) (Fig. 4D, E, beige).

An additional 20.7% ($n = 92$) of substrate-specific variants exhibited reduced catalytic activity toward the primary substrate, but were similarly stable or slightly more stable than WT (average expression score: 1.00 ± 0.06 ; WT score: 1). These mutations tended to occur closer to the active site (average distance: $7.96 \pm 6.61 \text{ \AA}$) and reflect a trade-off in which enhanced specificity is achieved at the cost of catalytic efficiency (Fig. 4D, E, gray).

The remaining 54.4% of variants ($n = 242$) increased both specificity and catalytic activity but imposed the most substantial reduction in stability, with the lowest expression scores (0.87 ± 0.08) and the greatest average distance from the active site (average distance: $17.1 \pm 8.81 \text{ \AA}$) (Fig. 4D, E, colored). These results highlight how functional gains can often originate from distal mutations, even when accompanied by destabilizing effects. Notably, the most impactful specificity-conferring mutations clustered near the active site (Fig. 3C), suggesting that large shifts in substrate preference often involve functional compromises, passing through regions of reduced fitness along the evolutionary trajectory.

Single mutations reshape substrate preference via distinct structural mechanisms

To understand in more detail the structural mechanisms underlying substrate specificity, we characterized the catalytic properties of 30 single-mutant DAOx variants chosen for their high Sp-scores. Each variant, along with WT-DAOx was expressed on yeast and assayed for Michaelis–Menten kinetic parameters against all five substrates. We determined V_{max} and K_{M} values and compared them to WT values to assess changes in substrate affinity and catalytic rate (Supplementary Table 5).

Among D-Ala-selective variants, F58W and F58Y exhibited the highest specificity. Located at the entrance of the substrate tunnel, these mutations increase steric hindrance, restricting access to larger substrates. While WT-DAOx already contains a bulky phenylalanine at this position, introducing even larger residues (Trp or Tyr) significantly shifted the enzyme's preference toward D-Ala. In the F58W variant, the D-factor for D-Ala over D-Phe increased by >200 -fold ($D_{\text{D-Ala, D-Phe}}$: F58W = 42.9, WT = 0.2), while an average ~ 20 -fold increase in D-factor was observed in F58W for D-Ala over D-Met and D-Asn (Fig. 5 i, Supplementary Table 5).

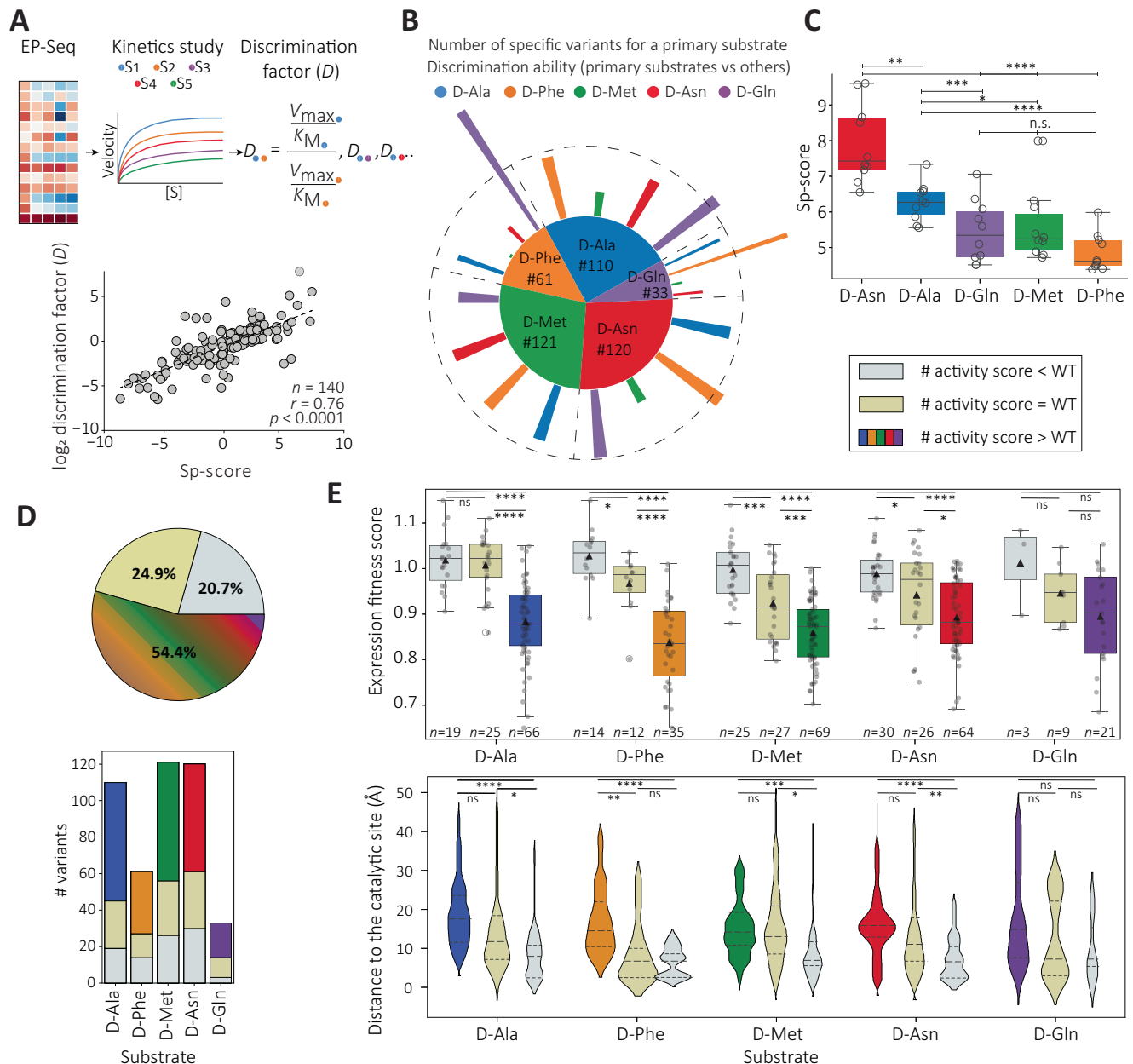


Fig. 4 | Validation and global analysis of substrate specificity scores.

A Comparison of EP-Seq-derived specificity scores (Sp-scores) with single-clone discrimination factors (D) determined by Michaelis–Menten kinetic analysis across 140 variant–substrate pairs. Dashed line: linear regression; r : Pearson correlation coefficients, n : number of datapoints, p : p -values two-tailed. **B** Pie chart showing the number of specificity-conferring unique mutations identified for each substrate. Outer bar heights represent the intensity of discrimination of variants specific for the substrate shown in the pie chart against each of the non-cognate substrates. **C** Box plot showing Sp-score distributions for the 10 most selective variants per substrate. Center lines indicate medians, boxes the interquartile range (25th–75th percentiles), whiskers the minimum and maximum values. Two-sided unpaired t -tests were performed for all pairwise comparisons. **D** Pie chart (top) summarizing different functional categories of substrate-specific variants: those with reduced (gray), unchanged (beige) or increased (multi-colored) catalytic activity toward

their cognate substrate. The bar plot shows the distribution of these categories within each of the five substrate datasets. **E** (Top) Box plots of expression fitness scores (proxy for stability) for each substrate-specific variant set grouped by catalytic activity relative to WT. The inner line indicates the median, the box spans the 25th to 75th percentiles, whiskers extend to the most extreme values within 1.5 times the box range, and the mean is shown as a marker. (Bottom) Violin plots showing the distribution of distances from each mutation site to the catalytic center grouped by substrate and catalytic activity relative to WT. Inner lines indicate the median and interquartile range. n.s. not significant. * indicates $p < 0.05$, ** indicates $p < 0.01$, *** indicates $p < 0.001$ and **** indicates $p < 0.0001$. Exact p -values for all statistical comparisons are reported in the Source Data file. Two-sided unpaired t tests were performed for all pairwise comparisons. n indicates the number of DAOx variants scores in each group. Source data are provided as a Source Data file.

For the D-Phe substrate, N54L emerged as the most selective variant. In the WT-DAOx catalytic site, Asn54 coordinates a water molecule²⁸ and orients substrates through its polar side chain that projects toward the center of the catalytic pocket. Replacing Asn54 with hydrophobic Leu without increasing steric bulk likely alters local

solvent interactions, enhancing binding of large hydrophobic substrates. This enhances compatibility with D-Phe and D-Met over the smaller D-Ala (D_{D-Phe}, D_{D-Ala}: N54L = 22.9, WT = 5.1). The structurally similar N54I showed a comparable selectivity profile as N54L (D_{D-Phe}, D_{D-Ala}: N54I = 23.3, WT = 5.1) (Fig. 5 ii, Supplementary Table 5).

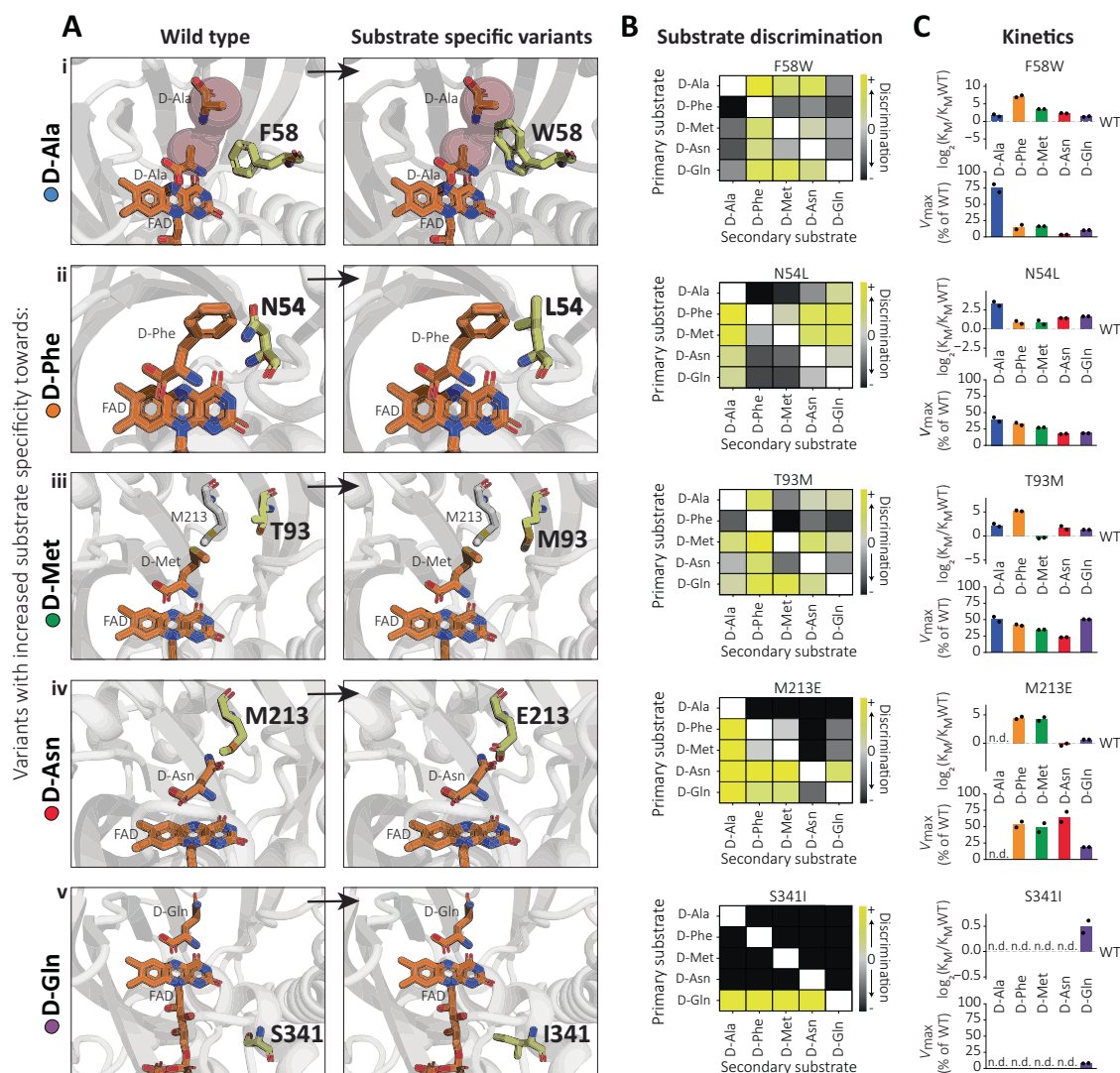


Fig. 5 | Structural and kinetic insights into substrate-specific DAOx variants.

A Top substrate-selective mutations for each of the five tested substrates mapped onto the three-dimensional structure of DAOx (PDB ID: 1COP). The FAD cofactor and D-amino acid substrates are shown as orange sticks; oxygen, nitrogen, and sulfur atoms are colored red, blue, and yellow, respectively. When relevant, the substrate access tunnel is depicted as light red spheres⁵⁵. Wild-type or substituted residues at the mutation sites are shown in yellow sticks; all other residues are shown in gray. **B** Heatmaps of substrate discrimination factor for all pairwise combinations of primary (y-axis) and secondary (x-axis) substrates, calculated as

the catalytic efficiency (V_{max}/K_M) for the primary substrate divided by that for the secondary substrate. **C** Catalytic profiles of selected mutants expressed on the surface of the yeast. Relative changes in K_M and V_{max} values compared to the wild-type enzyme for each of the five substrates, illustrating how specificity is tuned through alterations in substrate affinity and catalytic turnover. n.d. (not detected) indicates no measurable activity, with signal below the detection limit of the assay. $n = 3$ independently measured samples from the same cell preparation. Source data are provided as a Source Data file.

On D-Met, the T93M mutation located just outside the active site increased selectivity by modestly improving D-Met affinity while broadly reducing catalytic efficiency on the other substrates. A similar effect was observed in T56M, likely due to stabilization of nearby contacts involving Met213. For T93M, discrimination between D-Met and D-Phe increased ~40-fold ($D_{D-Met, D-Phe}$: T93M = 39.8, WT-DAOx = 0.9.) (Fig. 5 iii, Supplementary Table 5).

For the polar substrate D-Asn, the charged substitution M213E conferred strong selectivity, consistent with prior reports implicating this position in electrostatic tuning of the active site^{41,42}. M213E reduced overall enzyme activity but preserved or enhanced affinity for D-Asn and D-Gln, with absolute selectivity against D-Ala and an average ~30-fold increase in D-factor for D-Asn over the other two hydrophobic substrates D-Phe and D-Met ($D_{D-Asn, D-Phe}$: M213E = 0.7, WT-DAOx = 0.02; $D_{D-Asn, D-Met}$: M213E = 0.73, WT-DAOx = 0.02). The similar variant M213D further reinforced these observations (Supplementary Table 5).

For the polar substrate D-Gln, we found that the variant S341I exhibited striking selectivity despite the mutation being >8 Å away from the catalytic site. The mutation at position S341 can alter the orientation of nearby residues Y338 and Q339, which are directly involved in substrate coordination. While K_M for D-Gln slightly increased, V_{max} dropped to ~10% of WT. No measurable activity was observed for any of the other four substrates, even under extended assay conditions, making this variant fully selective for D-Gln, which is notably one the least favored substrates of WT-DAOx. This highlights how even distal mutations can fine-tune specificity through subtle geometric or allosteric effects (Fig. 5 v, Supplementary Table 5).

To reinforce the validity of our observations, we selected the five DAOx variants that exhibited the most pronounced shifts in substrate specificity toward each of the five tested substrates (F58W, N54L, T93M, M213E, S341I) for soluble expression in *E. coli* and subsequent purification. These variants, together with soluble WT-DAOx, were

comprehensively characterized for their Michaelis–Menten kinetics against each of the five substrates used in this study (Table 1, Supplementary Fig. 8). We then calculated discrimination factors and compared them to the corresponding profiles obtained from the yeast-displayed formats (Supplementary Fig. 9). For all unique substrate pairs across these five variants, the determined D-factors showed strong agreement between the soluble and yeast-displayed forms ($n = 36$, $r = 0.87$, $p < 0.0001$; Supplementary Fig. 9A). All variants retained the specificity profiles observed in the yeast-displayed format. Notably, S341I maintained absolute selectivity for D-Gln, even when tested at concentrations 20-fold higher than WT and under extended incubation times (Supplementary Fig. 9B).

Distal mutations as modulators of substrate specificity

Distal mutations can significantly influence enzyme activity and substrate specificity, although their identification and mechanistic roles are challenging to detect. These effects are often mediated through conformational or allosteric changes and are highly context-dependent^{43–45}. Previous deep mutational scanning studies have reported contrasting patterns. In some cases, specificity-determining mutations are mainly distal ($>9 \text{ \AA}$ from the catalytic site)⁴⁸, while in others, they cluster within the first shell residues around the active site^{15,17,46}.

In DAOx, we found that the strongest specificity-shifting mutations were predominantly located within $8 \text{ \AA}$ of the catalytic site (Fig. 3C), yet these accounted for only 30% ($n = 101$) of all such mutations. The remaining 70% ($n = 231$) were scattered across more distal regions of the enzyme. Notably, we identified a hotspot along α -helix 10 (residues 248–263), where positions 254, 258, and 262 were among the most frequently mutated positions affecting specificity (Fig. 6A). These residues occupy alternating helical turns and project their side chains in the same direction, forming interactions with β -strands 9–11 (residues 212–235). This region shapes the catalytic cleft and contains critical substrate-coordinating residues, including M213, S215, and P221 (Fig. 6B) which are among the most important positions for substrate specificity across all tested substrates (Fig. 3D, Supplementary Fig. 7).

For these distal positions, we characterized two variants per site at the single clone level. At positions 254 and 258, substitutions such as R254C/D and H258W/R resulted in a twofold increase in substrate selectivity toward the bulky hydrophobic substrate D-Phe over all other tested substrates. In contrast, substitutions at position L262 (L262V/A), which introduce smaller hydrophobic side chains, led to a twofold increase in specificity for the smaller substrate D-Ala over all other substrates, with the exception of D-Gln (Fig. 6C). These results show how distal mutations outside the catalytic site can reshape the protein’s structural network, subtly modifying the catalytic pocket’s geometry and influencing substrate preference. The close proximity of α -helix 10 to the dimer interface hints that oligomerization-induced structural reorganization can further modulate the catalytic site and contribute to substrate specificity.

Combinatorial mutations enable control over substrate specificity

To further refine substrate specificity for engineering applications, we tested whether combining substrate-specific mutations would produce additive or synergistic effects on substrate preference (Fig. 7). Since the most selective single-point mutations were clustered within the catalytic site and also enhanced structural stability, we hypothesized that combining these mutations could further sharpen substrate specificity without significantly compromising stability or expression levels.

We constructed six double mutants and measured their kinetic parameters across all five substrates, comparing the specificity profiles to those of the corresponding single mutants (Fig. 7A, B). For example,

Table 1 | Kinetic parameters of soluble WT- and substrate-specific-DAOx variants tested with each of the five substrates used in this study

DAOx soluble variants	D-alanine		D-phenylalanine		D-methionine		D-asparagine		D-glutamine	
	K_M [mM] (%WT)	k_{cat} s ^{−1} (%WT)	K_M [mM] (%WT)	k_{cat} s ^{−1} (%WT)	K_M [mM] (%WT)	k_{cat} s ^{−1} (%WT)	K_M [mM] (%WT)	k_{cat} s ^{−1} (%WT)	K_M [mM] (%WT)	k_{cat} s ^{−1} (%WT)
WT	5.62 ± 0.81 (100)	1.47 ± 0.06 (100)	2.56 ± 0.35 (100)	1.93 ± 0.05 (100)	2.02 ± 0.16 (100)	1.37 ± 0.05 (100)	47.06 ± 0.81 (100)	0.84 ± 0.11 (100)	18.84 ± 1.99 (100)	0.81 ± 0.05 (100)
F58W	10.43 ± 1.33 (190)	2.21 ± 0.20 (150)	43.73 ± 7.55 (1750)	0.22 ± 0.03 (11)	8.90 ± 0.72 (445)	0.43 ± 0.00 (31)	47.90 ± 9.34 (102)	0.06 ± 0.01 (7)	30.70 ± 1.57 (165)	0.07 ± 0.00 (9)
N54L	22.04 ± 2.52 (400)	1.20 ± 0.17 (82)	1.60 ± 0.04 (63)	1.07 ± 0.02 (56)	1.35 ± 0.12 (68)	0.90 ± 0.02 (65)	59.29 ± 8.65 (126)	0.32 ± 0.01 (39)	36.75 ± 1.61 (197)	0.21 ± 0.02 (26)
T93M	16.94 ± 2.41 (302)	0.15 ± 0.03 (10)	13.75 ± 0.26 (543)	0.08 ± 0.00 (4)	0.74 ± 0.01 (37)	0.12 ± 0.00 (8)	41.02 ± 14.05 (87)	0.03 ± 0.01 (4)	20.86 ± 1.10 (111)	0.06 ± 0.00 (7)
M213E	29.71 ± 15.59 (518)	0.13 ± 0.06 (9)	21.52 ± 1.04 (847)	1.85 ± 0.09 (96)	9.99 ± 1.3 (499)	1.75 ± 0.06 (127)	17.26 ± 0.98 (37)	1.30 ± 0.09 (156)	13.09 ± 0.86 (70)	0.22 ± 0.01 (27)
S341I	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	9.53 ± 0.25 (51)	0.02 ± 0.00 (2)

Values in parentheses indicate K_M and k_{cat} as a percentage of the corresponding value for WT-DAOx. n.d. (not detected) indicates no measurable activity, with signal below the detection limit of the assay. Values are shown as the mean ± standard deviation from $n = 3$ three independently measured samples from the enzyme preparation.

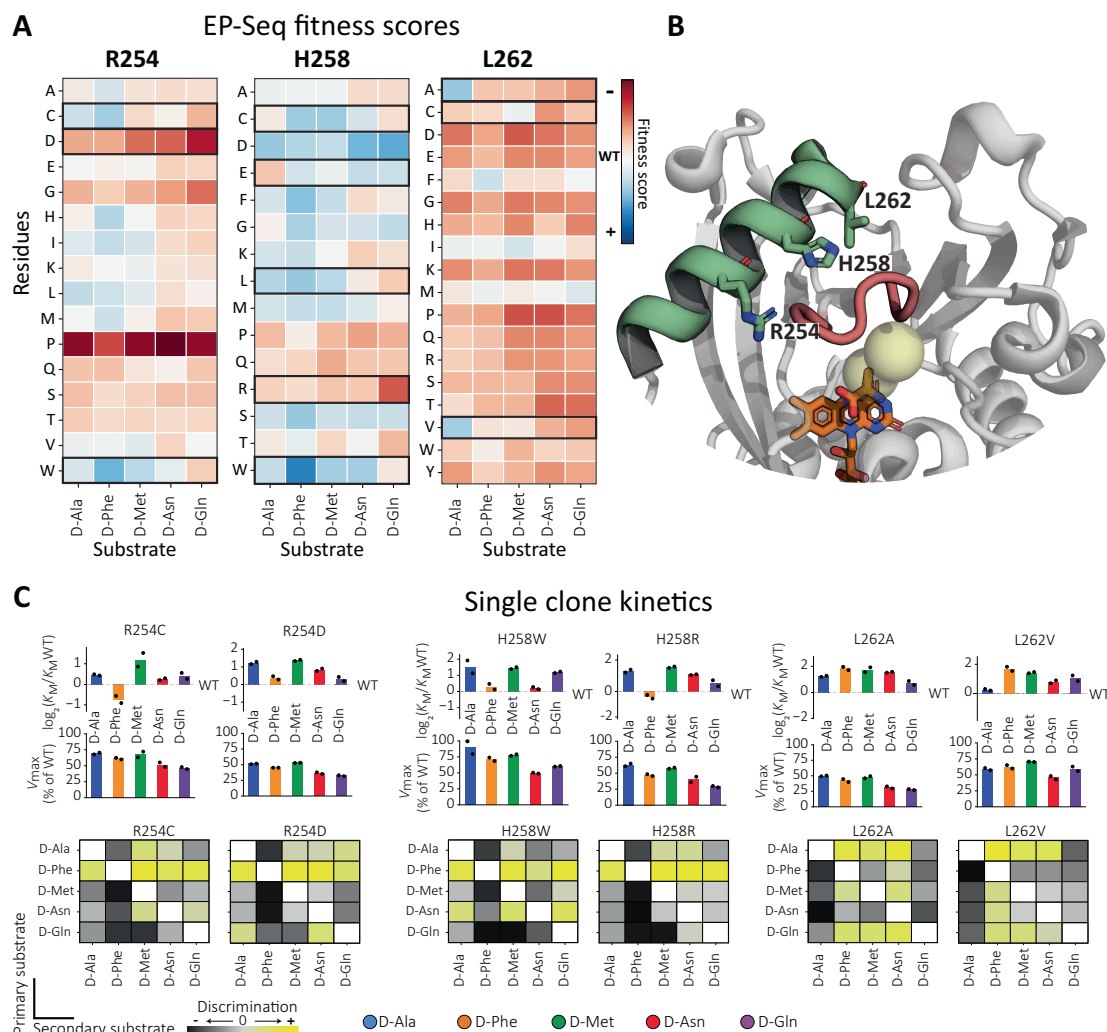


Fig. 6 | α -Helix 10 as an allosteric site modulating substrate specificity in DAOx.

A Heatmaps of EP-Seq fitness scores for single-point mutants at positions 254, 258, and 262 across all five substrates (D-Ala, D-Phe, D-Met, D-Asn, and D-Gln), highlighting position-specific effects on substrate preference. **B** Structural visualization of positions 254, 258, and 262 on α -helix 10, showing side chains oriented toward β -strands 9–11 (residues 212–235). The loop between β -strands 10 and 11 is shown in red. The FAD cofactor and D-Ala are rendered as sticks (oxygen in red, nitrogen in blue). The substrate tunnel is indicated by light yellow spheres. This view illustrates

how distal mutations can alter catalytic site geometry through long-range interactions. **C** (Top) Kinetic analysis of six representative yeast displayed variants (two per position), showing changes in reaction velocity and substrate affinity across the five tested substrates. (Bottom) Heatmaps of substrate discrimination factor for all pairwise combinations of primary (y-axis) and secondary (x-axis) substrates, calculated as the catalytic efficiency ($V_{\max}/K_M$) for the primary substrate divided by that for the secondary substrate. Source data are provided as a Source Data file.

the F58W mutation that introduced strong steric hindrance at the substrate tunnel and consistently reduced activity toward D-Phe. When combined with mutations that favor D-Ala such as S215Y or A222M, we observed a sharp drop-in activity for all other substrates and a pronounced enhancement in D-Ala specificity. By contrast, combining F58W with M213E, which introduces a negative charge at the active site, abolished D-Ala activity and shifted specificity toward the polar substrates D-Asn and D-Gln (Fig. 7C, top). The M213E/S215Y double mutant further intensified this specificity shift, redirecting enzyme activity almost entirely toward D-Asn and D-Gln, two substrates that are only minimally processed by WT-DAOx.

We also combined N54L and H258W, both of which favor D-Phe individually, and observed a strong additive effect: the double mutant retained over 75% catalytic efficiency for D-Phe while showing greatly reduced activity on all other substrates. However, not all combinations were additive. The T56H/T93M mutant, designed to enhance specificity on D-Met, maintained activity on multiple hydrophobic substrates without a clear preference, demonstrating that certain mutations can interfere with each other's effects (Fig. 7C, bottom).

Discussion

Substrate specificity lies at the intersection of an enzyme's chemical and biophysical characteristics. Once viewed as a static feature dictated solely by rigid active-site geometry, substrate specificity is now understood as an emergent property arising from conformational flexibility and long-range intramolecular contact networks. Enzymes are highly dynamic and sample multiple conformations governed by multiple weak interactions that collectively define substrate recognition^{44,45,47,48}. Despite these insights, encoding this complexity through enzyme design remains a major challenge due in large part to our limited understanding of how distant mutations exchange information and influence the catalytic geometry.

Some work has successfully enhanced specificity by designing rigid substrate-binding sites at the cost of overall activity, and later recovering activity back through directed evolution^{49–51}. But concrete principles derived from experimental data describing how specificity is achieved during evolution remain elusive. Here, we contribute to this understanding by presenting a comprehensive, single-mutation resolution map of substrate specificity in the D-amino acid oxidase from

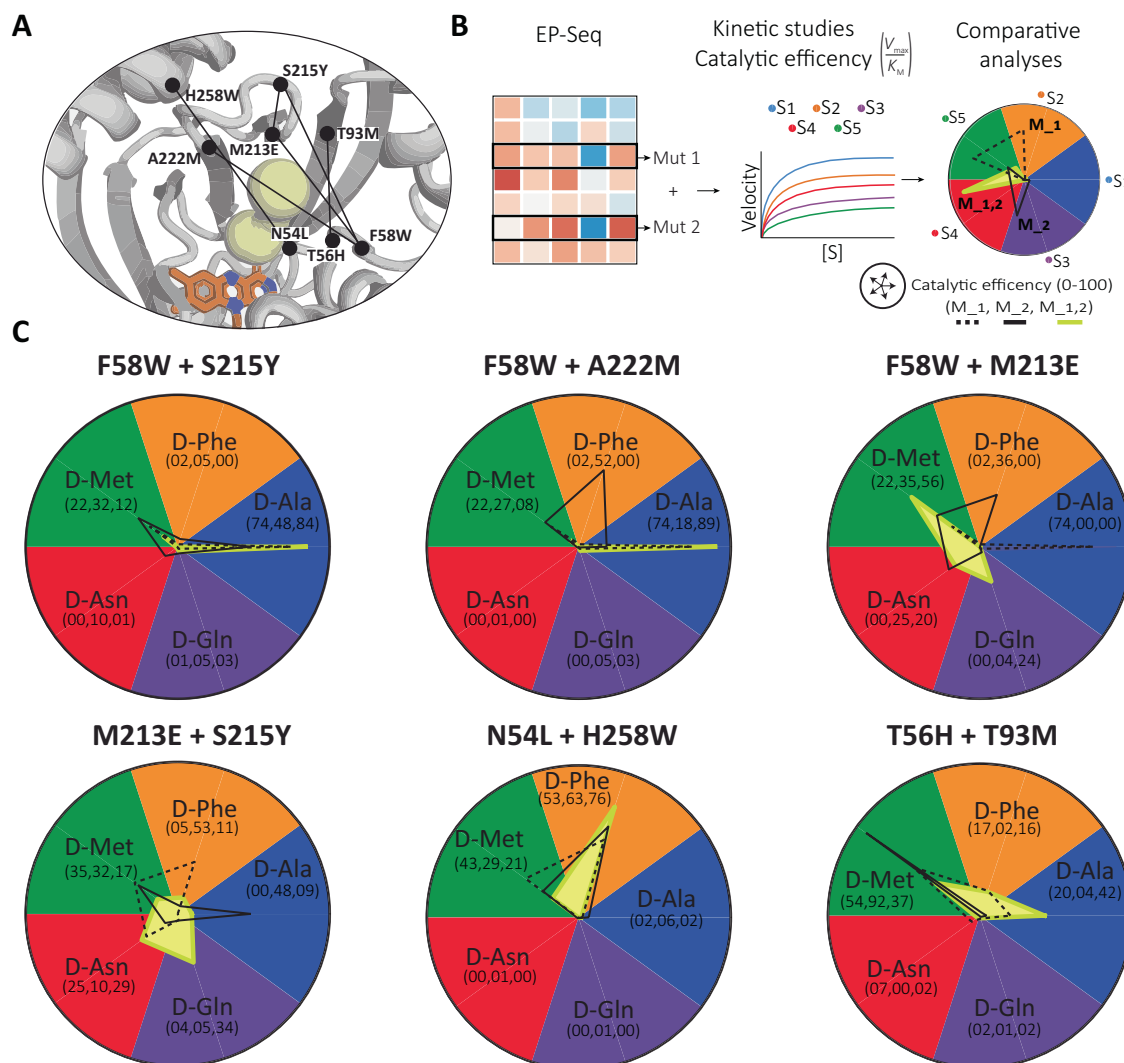


Fig. 7 | Rational engineering of substrate specificity through combinatorial mutations. **A** Structural mapping of single-mutant sites selected for double mutant design in DAOx (PDB ID: 1COP). Pairs were chosen based on complementary substrate specificity profiles. **B** Workflow illustrating the rational design process: substrate-specific single mutants identified by EP-Seq were first characterized individually and then combined into double mutants. The catalytic efficiency (V_{max}/K_M) of each double mutant was measured across all five substrates and compared to the corresponding single mutants to assess additive or synergistic effects on

substrate preference. **C** Radar plots summarizing the catalytic efficiency profiles of single and double mutants. For each substrate, the relative percentage of catalytic efficiency of the DAOx variants is indicated in parentheses, percentages were rounded to the nearest integer; therefore, totals may not sum to exactly 100%. Curves show the performance of single mutant 1 (dashed black line), single mutant 2 (solid black line), and the double mutant (solid yellow filled shapes), highlighting shifts in substrate specificity. Source data are provided as a Source Data file.

Rhodotorula gracilis. By leveraging our high-throughput EP-Seq method and exploiting the enzyme's natural promiscuity in accommodating chemically diverse substrates^{41,52}, we systematically mapped how the mutational fitness landscape shifts across substrates with distinct properties, offering insights into evolutionary routes to enzyme specificity.

Our results show that mutations near the catalytic site contribute most significantly to substrate specificity. However, these high-impact mutations often come at a substantial cost to catalytic activity. This supports a key evolutionary principle: natural enzymes can rapidly evolve sharp specificity via negative selection mechanisms (steric hindrance or electrostatic repulsion) as long as the functional cost is tolerable. However, other evolutionary routes toward specificity are also available. We identified hundreds of distal mutations that, although generally weaker in magnitude, tend to preserve catalytic activity better and allow specificity to evolve through neutral or slightly biased drift.

Beyond these evolutionary insights, our work establishes EP-Seq as a robust platform for high-throughput engineering of enzymes that

can be functionally displayed on the cell surface and whose activity can be coupled, directly or via a cascade, to peroxide production. The DMS-based Sp-score metric presented here shows a strong correlation with discrimination factors calculated from single-clone kinetic measurements, for both yeast-displayed and soluble variants. Among the strongest single mutations, variants such as F58W, and M213D/E shifted substrate preference by >200-fold. One variant (S341I) achieved exclusive selectivity for D-Gln, one of the WT enzyme's least favored substrates. We further show that combining mutations with complementary specificity effects can further refine substrate discrimination, offering a powerful strategy for engineering highly selective biocatalysts.

Methods

Expression and surface display of DAOx monoclonal populations and variant library

The pYDKan WT-DAOx plasmid and DAOx variant library used in this study were previously constructed and described¹⁹. The plasmids were

transformed into *Saccharomyces cerevisiae* strain EBY100 using the lithium acetate method⁵³, and positive colonies were selected on synthetic defined (SD) plates or liquid medium containing 2% (w/v) glucose and lacking tryptophan (-Trp). As previously described¹⁹, to induce expression and surface display of the Aga2-DAOx fusion proteins, positive transformants were first cultivated in SD -Trp liquid medium with 2% (w/v) glucose at 30 °C for 24 h with shaking at 200 rpm, until reaching an OD₆₀₀ of approximately 8. Cells were then transferred to fresh SD -Trp induction medium containing 0.2% (w/v) glucose, 1.8% (w/v) galactose, and 100 mM citrate/phosphate buffer (pH 7.0) and incubated at 20 °C for 48 h. Following induction, yeast cells were pelleted, washed with PBS containing 0.1% (w/v) bovine serum albumin (BSA), and labeled with antibodies targeting the C-terminal 6×His tag to assess surface display of the fusion proteins.

Yeast antibody staining to detect surface-displayed DAOx

As outlined in our previous work¹⁹, following induction of protein expression and surface display, yeast cells were washed twice with PBS supplemented with 0.1% (w/v) BSA. Cells were then resuspended at a density of 2×10^6 cells per 100 μ l in the same buffer containing a 1:500 dilution of mouse monoclonal anti-6xHis antibodies (Thermo Fisher Scientific, Cat# MA1-21315, RRID: AB_557403). Samples were incubated with the primary antibody for 30 minutes at room temperature, followed by two additional washes with PBS/BSA. Next, cells were resuspended in PBS/BSA(0.1%) at the same concentration (2×10^6 cells/100 μ l) and incubated with a 1:500 dilution of Alexa Fluor™ 594-conjugated goat anti-mouse secondary antibody (Thermo Fisher Scientific, Cat# A-11005, RRID: AB_2534073) for 30 min at 4 °C. After incubation, samples were washed twice more and prepared for analysis. Flow cytometry was performed using an Attune NxT cytometer (Thermo Fisher Scientific) equipped with 488 nm and 561 nm lasers. Surface expression of DAOx was quantified by recording and analyzing 10,000 events per sample.

Soluble expression of WT-DAOx

The wild-type D-amino acid oxidase was expressed in *Escherichia coli*. The gene encoding the wild-type enzyme was cloned into the pET28 bacterial expression vector. Plasmids were transformed into BL21 (DE3) cells and plated on kanamycin-selective LB plates. From a single colony, a liquid LB kanamycin (LB Kan) culture (3 mL) was started and grown overnight at 37 °C with shaking (200 rpm). One milliliter of this overnight culture was used to inoculate 100 mL LB Kan and allowed to grow for 2.5 h at 37 °C to an OD₆₀₀ of 0.5. To induce expression, IPTG was added to a final concentration of 0.5 mM, and the culture was grown overnight for 18 h at 25 °C with shaking (200 rpm). The following day, cells were centrifuged at $4000 \times g$ for 15 min, and the supernatant was discarded. The pellet was placed on ice for 15 min and then resuspended in 10 mL of lysis buffer (50 mM TRIS, pH 8.0, 50 mM NaCl, 0.1% (v/v) Triton X-100, 5 mM MgCl₂). Cells were lysed by sonication for 10 min (Sonifier cell disruptor Branson Digital Sonifier, USA) and then centrifuged at $18,000 \times g$ for 20 min. The supernatant was loaded onto a gravitational flow column packed with 2 mL of HisPur Ni-NTA resin. Columns were washed with 6 column volumes of PBS buffer first and then 6 column volumes of PBS buffer containing 20 mM imidazole. Finally, the bound protein was eluted in 5 mL PBS buffer containing 500 mM imidazole. Elution fractions were concentrated using Vivaspins concentration columns (10 kDa cutoff). Subsequently, purity was increased, and residual salts from the affinity purification protocols were removed by size exclusion chromatography at a speed of 0.5 mL/min. The most abundant fraction was collected and the purity and molecular weight of the eluted protein were tested and confirmed by running 2 μ g of eluted protein through a 12% polyacrylamide SDS gel.

Kinetic characterization and comparison of soluble and yeast-displayed WT-DAOx

Kinetics measurements for soluble DAOx wild type were performed through Amplex Red activity assay by incubating the purified wild type enzyme with increasing concentrations of each tested substrate together with 5.6 μ M HRP and 100 μ M Amplex Red in PBS (pH 7.5). For measurement we used the following enzyme and substrate concentration. D-alanine, D-phenylalanine, D-methionine, D-leucine were tested at 0.27 mM, 0.55 mM, 1.1 mM, 2.2 mM, 4.4 mM, 8.8 mM, 17.5 mM and 35 mM with a WT-DAOx concentration of 0.058 nM. D-histidine was tested at 0.55 mM, 1.1 mM, 2.2 mM, 4.4 mM, 8.8 mM, 17.5 mM and 35 mM with a WT-DAOx concentration of 0.058 nM. D-serine was tested at 1.1 mM, 2.2 mM, 4.4 mM, 8.8 mM, 17.5 mM, 35 mM, 70 mM and 140 mM with a WT-DAOx concentration of 0.058 nM. D-Threonine was assayed at 1.1 mM, 2.2 mM, 4.4 mM, 8.8 mM, 17.5 mM, 35 mM, 70 mM and 140 mM with a WT-DAOx concentration of 0.29 nM. D-glutamine and D-asparagine were assayed at 0.4 mM, 0.8 mM, 1.6 mM, 3.2 mM, 6.3 mM, 12.5 mM, 25 mM and 50 mM with a WT-DAOx concentration of 0.29 nM. The fluorescence signal was read over time at 590 nm and converted to concentration of H₂O₂ using a calibration curve. The linear range of each reaction was used to extrapolate the reaction velocity and finally plotted versus substrate concentration to fit Michaelis–Menten model and extract K_M . k_{cat} was then calculated by dividing the measured V_{max} for each respective substrate by the concentration of enzyme used in the reaction.

Yeast cell populations positive for the display of DAOx (1'000'000 for D-Threonine D-glutamine and D-asparagine, 200'000 for D-alanine, D-phenylalanine, D-methionine, D-leucine, D-histidine and D-serine) were assayed for D-amino acid oxidase activity by mixing the cells with, 5.6 μ M HRP, and 100 μ M Amplex Red and increasing concentrations of each substrate in PBS (pH 7.5). D-alanine, D-phenylalanine, D-methionine, D-leucine were tested at 0.27 mM, 0.55 mM, 1.1 mM, 2.2 mM, 4.4 mM, 8.8 mM, 17.5 mM and 35 mM. D-histidine was tested at 0.55 mM, 1.1 mM, 2.2 mM, 4.4 mM, 8.8 mM, 17.5 mM. D-serine and D-threonine were tested at 1.1 mM, 2.2 mM, 4.4 mM, 8.8 mM, 17.5 mM, 35 mM, 70 mM and 140 mM. D-glutamine was assayed at 0.4 mM, 0.8 mM, 1.6 mM, 3.2 mM, 6.3 mM, 12.5 mM, 25 mM and 50 mM. D-asparagine was assayed at 0.5 mM, 1 mM, 2.2 mM, 4.4 mM, 8.8 mM, 17.5 mM, 35 mM and 70 mM. The fluorescence signal was read over time at 590 nm and converted to concentration of H₂O₂ using a calibration curve. The linear range of each reaction was used to extrapolate the reaction velocity and finally plotted versus substrate concentration to fit Michaelis–Menten function and extract K_M and V_{max} .

DAOx single cell tyramide/peroxidase proximity labeling assay

General procedure: the activity of yeast-displayed DAOx wild type was evaluated using a tyramide-based single-cell assay. Cells were diluted to a final concentration of 500 cells/ μ l and incubated with a 1:200 dilution of Alexa Fluor™ 488 Tyramide Reagent (Thermo Fisher Scientific, B40953), 55 μ M horseradish peroxidase (HRP; Sigma-Aldrich, 77332), and the appropriate concentration of D-amino acid substrate, adjusted according to the specific substrate and experimental application. Reactions were carried out at 25 °C for a defined period depending on the assay conditions. Following incubation, cells were pelleted by centrifugation at $13,000 \times g$ for 3 min using a benchtop centrifuge. The pellet was washed twice with PBS containing 0.1% (w/v) BSA and 0.05% TWEEN-20, then resuspended in the same buffer for flow cytometric analysis.

Time course tyramide assay: time course analyses of the tyramide assay were conducted under the same conditions as described above, with each substrate concentration set to its respective calculated K_M . Specifically, the concentrations were as follows: 7.3 mM D-alanine, 2 mM D-methionine, 1.5 mM D-phenylalanine, 14.4 mM D-glutamine, and 45.8 mM D-asparagine. For reactions utilizing D-alanine as a

substrate, incubation times were set at 5, 10, 20, 30, 40, 50, 60, 70, 80, and 90 min, respectively. Reactions involving D-phenylalanine and D-methionine were incubated for 1, 5, 10, 20, 30, 40, 50, 60, 70, 80, and 90 minutes, respectively. Lastly, reactions using D-asparagine and D-glutamine as substrates were incubated for 10, 20, 30, 40, 50, 60, 80, 100, 120, and 140 min, respectively.

DAOx variants library tyramide assay: catalytic activity of displayed DAOx variant enzymes was assayed through a single-cell tyramide/peroxidase proximity labeling method. After induction of protein expression and cell surface staining of the yeast population as described above, 25 M yeast cells were mixed at a concentration of 500 cells/ μ l with 1/200 dilution of Alexa fluorTM 488 Tyramide Reagent (Thermo Fisher, B40953), 56.8 μ M HRP (Sigma-Aldrich, 77332) and one of the five substrates at concentration equal to K_M in 1xPBS and 0.75% (w/v) sodium alginate. The reactions performed with D-alanine, D-phenylalanine and D-methionine were incubated for 60 min at 25 °C. The reactions performed with D-glutamine and D-asparagine were incubated for 120 min at 25 °C. Afterwards, the cells were spun down for 3 min at 13,000 $\times g$ in a table top centrifuge. The cell pellet was washed twice with PBS 0.1% (wt/vol) BSA + 0.05% (vol/vol) Tween 20 and then used for flow cytometry and single cell sorting experiments.

Activity level sorting

EBY100 yeast cells expressing DAOx enzyme variants were stained to assess surface expression and assayed for activity using the single-cell tyramide assay. Cells were subsequently sorted on a FACSMelody cell sorter (BD Biosciences) equipped with 488 nm and 561 nm lasers and a 100 μ m nozzle, following the protocol described in our previous study¹⁹ with minor modifications. Cells were sorted into pre-wetted 5 ml FACS tubes containing 0.5 ml of 2 $\times$ concentrated SD-Trp medium supplemented with 4% (w/v) glucose and 1% (w/v) BSA. Yeast cells were first gated for singleton events and then the population divided into four bins based on the level of green fluorescent signal. Bin 1 was designed in order to include 99% of the population of cells negative to display or displaying inactive DAOx variants, using as reference the fluorescence level of the non-displaying yeast populations. The remaining part of the population of cells was equally divided into three yeast sub-populations with increasing fluorescent signal. We performed the sorting of two independently assayed yeast populations sorting each time between 5 M and 12 M total yeast cells therefore covering the entire variability of the library of maximum 200'000 variants at least 25 fold. After each cycle of sorting, yeast cells part of the same gated population were merged into 50 ml falcon tubes and pelleted 10 min at 4000 $\times g$ in a table top centrifuge. Afterwards, the supernatant was discarded and the cell pellet resuspended in 10 ml -Trp medium with 2% glucose supplemented with 100 μ g/ml Pen Strep. After sorting, all the cell populations were grown for 48 h at 30 °C shaking at 200 rpm before being sampled into aliquots of 50 M cells each and stored at -80 °C in 25% (v/v) glycerol.

Preparation of DNA libraries for illumina sequencing and data analysis

As previously outlined¹⁹, ~50 million yeast cells per sorted population were used as starting material for the preparation of Illumina sequencing libraries. Frozen cell pellets stored at -80 °C were thawed and incubated at room temperature for 5 min, then centrifuged at 13,000 $\times g$ for 1 min in a benchtop centrifuge. The supernatant was discarded, and the cell pellet was resuspended in 250 μ l of miniprep resuspension solution (GeneJET Plasmid Miniprep Kit, Thermo Fisher) supplemented with 4 μ l of Zymolyase (5 U/ μ l). Samples were incubated at 37 °C for 2 h with shaking at 900 rpm to digest the yeast cell walls. Following enzymatic treatment, plasmid DNA was extracted according to the manufacturer's protocol for the GeneJET Plasmid Miniprep Kit. The purified plasmid DNA was eluted in 15 μ l of nuclease-free water. The region containing the 15 N unique molecular identifier (UMI) was

amplified by PCR using NEBNext Ultra II Q5 Master Mix (New England Biolabs). Each 50 μ l PCR reaction contained 25 μ l of master mix, 5 μ l of forward primer (1 μ M), 5 μ l of reverse primer (1 μ M), and 15 μ l of template DNA. Primers were designed to target the region of interest and to be compatible with Nextera indexing library preparation. To enhance sequence diversity at one end of the amplicon, four staggered forward primers were used in separate PCR reactions, and the resulting products were pooled for sequencing. PCR cycling conditions were as follows: initial denaturation at 98 °C for 30 s; 18 cycles of 98 °C for 10 s and 72 °C for 30 s; followed by a final extension at 72 °C for 2 min. Reactions were held at 4 °C until further processing. PCR products were verified by electrophoresis on a 1% (w/v) agarose gel and purified using a standard DNA gel extraction kit. DNA concentration was measured, and samples were further purified using the DNA Clean and Concentrator-5 kit (Zymo Research). Unique Nextera indexing sequences were then added, samples were pooled, and sequencing was performed on an Illumina NovaSeq 6000 platform. Illumina read counts per sample are detailed in Supplementary Table 2. Demultiplexed sequencing reads were processed using a computational pipeline to extract UMI sequences and align them against a look-up table. UMI sequences were mapped using the BMap alignment algorithm⁵⁴ with the reference sequence file (ill_ref.fa)

```
bbmap.sh in = *.fastq ref = ill_ref.fa out = *.sam
```

BMap-generated sam files were further processed using the C script pib.c (process Illumina barcodes). This step extracted the sequences and associated quality scores for each UMI mapped by BMap, saving the output in FASTQ format.

```
./pib *.sam> *.fq
```

Each mapped UMI was then aligned against the sequences stored in the look-up table (lut_m.tsv) using the C script rib.c (read Illumina barcodes). Each barcode was assigned one of the following tags based on the alignment and quality criteria: 0 – not found in the look-up table; 1 – found in the look-up table; 2 – read quality below Q20; 3 – UMI length different from 15 nucleotides. We applied a quality filter at Q20, tagging all UMIs associated with reads of lower quality with tag 2.

```
./rib -t lut_m.tsv -q 20 *.fq > *.tsv
```

Finally, UMI sequences with tag 1 were extracted, sorted alphabetically and grouped by identity.

```
grep^1 *.tsv |sort|uniq -c|sed -E 's/^*//; s/ /\t/'>t1sct_*.tsv
```

All Illumina read files were processed using the computational workflow described above. The UMI sequence data and counts from each bin (t1sct_Bin#.tsv) were subsequently merged into a single file using the Jupyter notebook ill_tag1_bins. The C programs employed in this analysis were compiled with GCC version 7.4.0 on a 32-bit processor. Raw sequencing reads are accessible via the NCBI Sequence Read Archive under the Bioproject accession number: PRJNA1289092. The scripts, look up table and output data are available at <https://doi.org/10.5281/zenodo.15846928>.

Fitness scores calculation

As in our previous work¹⁹, the number of sequencing reads corresponding to each enzyme variant (r_v) was converted into the number of sorted cells for that variant (c_v) per sorted bin using the following Eq. (1):

$$\frac{r_v}{r_{\text{tot}}} = \frac{c_v}{c_{\text{tot}}} \quad (1)$$

Here, r_{tot} represents the total number of Illumina reads in the bin, and c_{tot} is the total number of cells sorted into the same bin. The final activity scores for each variant and substrate were then calculated as the expected fluorescent intensity across all four bins in the experiment. We computed a weighted mean (β) of the cell counts (c_v), using the median fluorescence intensity of the yeast population in each bin (ω) as the weighting factor:

$$\beta = \frac{\sum_{i=1}^{\text{bin}} \omega_i \cdot c_{vi}}{\sum_{i=1}^{\text{bin}} c_{vi}} \quad (2)$$

The fitness score (F) per variant were finally calculated as follow:

$$F = \log_2 \left(\frac{\beta_v}{\beta_{\text{wt}}} \right) \quad (3)$$

where β_v represents the weighted mean expression score of the variant enzyme, and β_{wt} corresponds to the score of WT-DAOx. To calculate the final consensus fitness score (F_{fin}) for each variant, we computed a weighted mean of the individual fitness scores (F_v) obtained from each replicate assay. The weighting factor used was the number of cells (c_v) associated with the measured fitness in each experiment (Eq. (4)).

$$F_{\text{fin}} = \frac{\sum_{j=1}^{\text{rep}} F_{vj} \cdot c_{vj}}{\sum_{j=1}^{\text{rep}} c_{vj}} \quad (4)$$

Normalization of activity fitness scores by expression levels

To account for differences in enzyme display (i.e., expression) levels, and before comparative analyses, activity scores were exponentiated (i.e., inverted from $\log_2$ scale) and normalized to expression scores obtained from our previously published dataset¹⁹. For this procedure, we included only variants with expression levels exceeding the mean expression of nonsense variants plus one standard deviation, to prevent inflated normalized activity scores due to extremely low expression values.

Comparative analysis and Sp-score

To identify mutations altering substrate preference, we calculated z-scores for each missense variant across all pairwise substrate combinations using the formula (Eq. (5)):

$$z - \text{score} = \text{Sp} - \text{score} = \frac{X - \mu}{\sigma} \quad (5)$$

where X is the normalized activity fitness difference of the variant measured between two substrates, and μ and σ are the mean and standard deviation of fitness differences across all variants for the same substrate pair. This metric, termed substrate specificity score (Sp-score), captures both the magnitude and direction of substrate preference changes. Using synonymous mutations as controls, we set a threshold off $|\text{Sp}| \geq 3$ (three standard deviations) to identify substrate-specific mutations, excluding >99% of synonymous variants.

Quantification of substrate discrimination across substrate pairs

The intensity of discrimination for variants specific to a primary substrate against all non-cognate substrates presented in Fig. 4B was calculated in two steps. First, for each substrate pair, we computed the proportion of number of cognate substrate – specific variants (e.g., D-Ala vs D-Phe specific variants, $|\text{Sp-score}| \geq 3$) relative to the total number of specific variants for the primary substrate (total D-Ala specific variants, $|\text{Sp-score}| \geq 3$). This proportion was then multiplied by the mean Sp-score of these variants toward the primary substrate

within the same substrate pair. The resulting metric integrates both the prevalence and magnitude of substrate discrimination.

DAOx structural properties calculations from 3D structure

Coordinates of all atoms in DAOx residues were extracted from the PDB structure 1COP (PDB ID: 1COP). The minimum distance from each residue to the crystallized substrate D-Ala was calculated and used as a metric for proximity to the catalytic site in the analysis.

Validation of EP-Seq fitness scores

Eighteen single mutant DAOx variants were selected based on DMS scores spanning negative to positive values across D-Ala, D-Phe, D-Met, D-Asn, and D-Gln. Genes encoding S48C, K109L, K109W, K109Y, D110I, D110M, D110W, L153R, A187E, A187K, G199E, Q200W, S215A, S215R, T237M, Y238W, S268Y and L310P were synthesized and cloned into the pYDKan plasmid in-frame with the Aga2p coding sequence using a BioXP 3250 synthetic biology workstation (Telesis Bio) or generated via PCR and KLD cloning to introduce mutations into the wild-type DAOx sequence. All sequences were codon-optimized for expression in *Saccharomyces cerevisiae* and verified by Sanger sequencing. Final plasmids were transformed into *S. cerevisiae* strain EBY100 using the lithium acetate method⁵³. Positive colonies were selected on synthetic defined (SD) agar plates (-Trp) with 2% (wt/vol) glucose and cultivated in -Trp liquid medium with 2% (wt/vol) glucose for 24 h at 30 °C to an $\text{OD}_{600} \sim 8$ with continuous shaking (200 rpm). Expression and display of Aga2-DAOx wild-type and mutant constructs were induced by transferring cells at $\text{OD}_{600} = 0.4$ to fresh -Trp liquid medium containing 0.2% (wt/vol) glucose, 1.8% (wt/vol) galactose, and 100 mM citrate/phosphate buffer (pH 7). Cells were grown for 48 h at 20 °C, then pelleted, washed with PBS containing 0.1% (wt/vol) BSA, and used for antibody labeling to detect the C-terminal 6×His tag of the displayed fusion protein and confirm expression. The cell populations expressing DAOx wild-type and selected variants were then assayed for activity using the Amplex Red method and the five substrates from the EP-Seq screen (D-Ala, D-Phe, D-Met, D-Asn, D-Gln). A total of 250,000 yeast cells were mixed with 5.6 μM HRP and 100 μM Amplex Red in PBS (pH 7.5), and each substrate was added at the same final concentration used in the EP-Seq workflow (7.3 mM D-Ala, 1.5 mM D-Phe, 2 mM D-Met, 46 mM D-Asn, 14 mM D-Gln). Fluorescence was measured at 590 nm every 50–60 s for at least 60 min and at a temperature of 27–30 °C. The linear range of the reaction for all enzyme variants and substrates was determined between 0 and 15 min, and the reaction slope was calculated. The single-clone activity score for each variant was obtained by dividing the reaction slope of the variant by that of the wild-type enzyme. Final scores were averaged from three independent measurements per cell population and plotted against the corresponding EP-Seq activity scores.

Kinetic study on yeast displayed DAOx substrate-specific variants

Based on our EP-Seq comparative analysis, we selected DAOx variants with the highest z-scores and specificity towards each of the tested substrates. Gene-encoding DAOx variants (Supplementary Table 5) were generated via PCR and KLD cloning to introduce mutations into the wild-type DAOx sequence. All sequences were codon-optimized for expression in *Saccharomyces cerevisiae* and verified by Sanger sequencing. Final plasmids were transformed into *S. cerevisiae* strain EBY100 using the lithium acetate method⁵³. Positive colonies were selected on synthetic defined (SD) agar plates (-Trp) with 2% (wt/vol) glucose and cultivated in -Trp liquid medium with 2% (wt/vol) glucose for 24 h at 30 °C to an $\text{OD}_{600} \sim 8$ with continuous shaking (200 rpm). Expression and display of Aga2-DAOx wild-type and mutant constructs were induced by transferring cells at $\text{OD}_{600} = 0.4$ to fresh -Trp liquid

medium containing 0.2% (wt/vol) glucose, 1.8% (wt/vol) galactose, and 100 mM citrate/phosphate buffer (pH 7). Cells were grown for 48 h at 20 °C, then pelleted, washed with PBS containing 0.1% (wt/vol) BSA, and used for antibody labeling to detect the C-terminal 6×His tag of the displayed fusion protein. Yeast cells expressing DAOx wild-type and mutant variants were analyzed by flow cytometry.

Yeast populations were first gated for single events, then divided into four gates based on Alexa Fluor™ 594 fluorescence to cover the entire fluorescence range. A total of 10,000 single cells from each population were analyzed. The median fluorescence value and percentage of cells in each gate were recorded. Each population was assayed in duplicate. A weighted mean (β) of the cell distribution (%c) was calculated, with the median fluorescence of each gate (ω) as the weighting factor (Eq. (6)).

$$\beta_{\text{sing, clone}} = \frac{\sum_{i=1}^{\text{gate}} \omega_i \cdot \%c_i}{\sum_{i=1}^{\text{gate}} \%c_i} \quad (6)$$

Finally, the single-clone expression score ($F_{\text{sing, clone}}$) was calculated by dividing the weighted mean expression of each DAOx variant (β_v) by the weighted mean expression of the wild-type DAOx (β_{wt}), both assayed using the same procedure (Eq. (7)).

$$F_{\text{sing, clone}} = \frac{\beta_v}{\beta_{\text{wt}}} \quad (7)$$

Yeast populations expressing variants were used to determine the kinetic parameters K_M and V_{max} of each DAOx mutant towards the five substrates D-Ala, D-Phe, D-Met, D-Asn, D-Gln. The number of cells used for each kinetic measurement was fixed to 400,000 and calculated by measuring the OD of each population using as reference value 1 OD₆₀₀ corresponding to 10 M cells in 1 ml. By maintaining fixed the number of cells and taking into account the expression level of each variant enzyme we wanted to allow not only relative comparative analysis towards substrates for the same variant but also compare the performance of the DAOx single mutants to the wild type enzyme. Each yeast population was assayed for D-amino acid oxidase activity by mixing the cells with 5.6 μM HRP, 100 μM Amplex Red and increasing concentrations of each substrate in PBS (pH 7.5) at room temperature (25–30 °C). Specifically, D-alanine, D-phenylalanine, D-methionine, were tested at 0.3125 mM, 0.625 mM, 1.25 mM, 2.5 mM, 5 mM, 10 mM, 20 mM and 40 mM, while D-asparagine and D-glutamine were assayed at 1.5625 mM, 3.125 mM, 6.25 mM, 12.5 mM, 25 mM, 50 mM, 100 mM, 200 mM. The fluorescence signal was read for at least 20 min at 590 nm. The linear range of each reaction was determined and used to extrapolate the reaction velocity and finally plotted versus substrate concentration to fit Michaelis-Menten function and extract K_M and V_{max} . Reaction velocities were determined for at least two independent replicate reactions at six or more substrate concentrations. These data were used to fit the Michaelis-Menten kinetic model. Due to a technical issue affecting one of the replicate reactions, the R254C variant tested with D-Ala was assessed using a single reaction only. Fitting results were considered reliable and accepted if the coefficient of determination (R^2) was >0.90 or higher. In cases where enzyme activity was too low to be accurately detected and the data did not meet the criteria for Michaelis-Menten fitting, reactions were repeated using longer incubation times and/or increased enzyme concentrations. If no improvement was observed, kinetic parameters (K_M and V_{max}) were classified as not detected (n.d.).

Discrimination factor calculation

The selectivity (D) of an enzyme for a primary substrate is defined as the ratio of the k_{cat}/K_M values for the primary substrate over the same

ratio for the secondary substrate. This can be expressed as (Eq. (8)):

$$D = \frac{(k_{\text{cat}}/K_M)_{\text{sub}_1}}{(k_{\text{cat}}/K_M)_{\text{sub}_2}} \quad (8)$$

In this manuscript, when yeast surface-displayed enzymes were used to measure catalytic performance, and the absolute enzyme concentration in the reaction could not be determined, we define the discrimination factor (D) between two substrates as the ratio of V_{max} and K_M (Eq. (9))

$$D = \frac{(V_{\text{max}}/K_M)_{\text{sub}_1}}{(V_{\text{max}}/K_M)_{\text{sub}_2}} \quad (9)$$

This relationship allows to calculate reliable discrimination factor values towards different substrates that reflect the catalytic properties of the soluble counterparts. For each variant we first normalize the V_{max} value to the single clone expression score ($F_{\text{sing, clone}}$) in order to remove the effect of different levels of expression from the detected reaction velocity. Then we calculated the ratio between V_{max}/K_M for all of the 5 substrates. And the discrimination factor (D) for all possible substrate combinations.

Soluble expression of WT and substrate-specific DAOx variants

DAOx variants (Table 1) were generated by site-directed mutagenesis using PCR and KLD cloning to introduce specific mutations into the wild-type DAOx gene, which was previously cloned into the pET28 expression vector. All constructs were verified by Sanger sequencing and transformed into *E. coli* NiCo21(DE3) for subsequent expression and purification. Cultures were grown in super optimal broth (SOB, 20 g/L tryptone, 5 g/L yeast extract, 2.5 mM KCl, 10 mM MgCl₂, MgSO₄, 10 mM NaCl, pH 7.0) at 37 °C with shaking until reaching an O.D.₆₀₀ of 0.6–0.8. The cultures were then induced using 1 mM isopropyl β-D-1-thiogalactopyranoside (IPTG) at 16 °C, with shaking, for 20 h. After harvesting via centrifugation, the cell pellets were stored at –80 °C. When needed, the cell pellets were thawed and resuspended at 4 °C in BugBuster® protein extraction reagent (MilliporeSigma, Burlington, MA, USA) according to the manufacturers protocol. The resulting cell lysate then underwent centrifugation at 30,000 × g for 30 min and was passed through a 0.45 μm filter. The clarified lysate was then supplemented with 0.1% (w/v) Tween 20 and 10 mM Imidazole, and incubated with Ni-NTA Resin (HisPur™, Thermo Fisher Scientific, Massachusetts, United States) at 4 °C, with shaking, for 1 h. Ni-NTA Resin was reconstituted by centrifugation 800 × g for 5 min and washed three times with 0.1% (w/v) Tween 20 and 10 mM Imidazole supplemented phosphate-buffered saline (PBS, 137 mM NaCl, 2.7 mM KCl, 10 mM Na₂HPO₄, and 1.8 mM KH₂PO₄, pH 7.4). The Ni-NTA Resin was then transferred to gravity flow columns (Pierce™, Thermo Fischer Scientific, Massachusetts, United States) and washed using an imidazole gradient, 20–250 mM imidazole-supplemented PBS. Protein-containing elution fractions were then concentrated in centrifugal filters (Amicon®, Merck, Darmstadt, Germany) and further processed using size-exclusion chromatography (SEC, Superose 12 10/300 GL column, Cytiva, Marlborough, United States) into PBS. SEC elution fractions containing the protein of interest were again concentrated using centrifugal filters and stored at –80 °C, until used in experiments. Protein concentrations were measured by spectrophotometry at 280 nm (NanoDrop 1000, 6 Thermo Scientific, DE, USA).

Kinetic study of soluble WT and substrate-specific DAOx variants

Soluble WT-DAOx and its variants (Table 1) were used to determine the kinetic parameters K_M and k_{cat} toward five substrates: D-Ala, D-Phe, D-Met, D-Asn, and D-Gln. Each enzyme variant was assayed for D-amino acid oxidase activity by mixing the cells with 5.6 μM HRP, 100 μM Amplex Red, and increasing concentrations of each substrate in PBS

(pH 7.5) at room temperature (25–30 °C). WT-DAOx was tested at a concentration of 0.25 nM, while the variants were tested at the following concentrations: N54L at 2 nM, F58W at 2 nM, T93M at 5 nM, M213E at 1 nM, and S341I at 5 nM. For each variant, D-Ala, D-Phe and D-Met were tested at concentrations of 0.3125, 0.625, 1.25, 2.5, 5, 10, 20, and 40 mM, while D-Asn and D-Gln were assayed at 1.5625, 3.125, 6.25, 12.5, 25, 50, 100, and 200 mM. The fluorescence signal was read for at least 20 min at 590 nm. The linear range of each reaction was determined and used to extrapolate the reaction velocity and finally plotted versus substrate concentration to fit Michaelis-Menten function and extract K_M and V_{max} . Reaction velocities were determined for at least three independent replicate reactions at six or more substrate concentrations. These data were used to fit the Michaelis-Menten kinetic model. Fitting results were considered reliable and accepted if the coefficient of determination (R^2) was ≥ 0.90 or higher. In cases where enzyme activity was too low to be accurately detected and the data did not meet the criteria for Michaelis-Menten fitting, reactions were repeated using longer incubation times and/or increased enzyme concentrations. If no improvement was observed, kinetic parameters (K_M and k_{cat}) were classified as not detected (n.d.).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data required to replicate this study and generated in this study are publicly available. DNA sequencing reads can be accessed through the NCBI Sequence Read Archive under BioProject accession [PRJNA1289092](https://doi.org/10.5281/zenodo.15846928). Raw data for all figures, and processed output data are available at <https://doi.org/10.5281/zenodo.15846928>. The PDB code of the previously published structure used in this study is [1COP](https://doi.org/10.5281/zenodo.15846928). Source Data are provided as a Source Data file. Source data are provided with this paper.

Code availability

All custom code and scripts used in this study are publicly available in the Zenodo repository at <https://doi.org/10.5281/zenodo.15846928>

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Author contributions

R.V. and M.A.N. conceived the study and drafted the manuscript. R.V. carried out the practical work and computational analyses. S.B. optimized and performed the expression and purification of the soluble DAOx variants. C.K. contributed to the conceptualization and optimization of the workflow. M.A.N. secured funding and administered the project.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Rosario Vanella or Michael A. Nash.

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